Feature Engineering Basics

1. Why features matter

Feature engineering is the process of turning raw data into inputs that make a machine learning model work better. A simple model with well-designed features often beats a complex model fed with raw values. Good features expose patterns that the algorithm cannot easily discover on its own.

2. Scaling numeric features

Many algorithms, including k-nearest neighbours, support vector machines and gradient-based neural networks, are sensitive to the scale of features. Min-max scaling maps a value x to (x - min) / (max - min), producing values between 0 and 1. Standardization maps x to (x - mean) / standard deviation, producing values with mean 0 and standard deviation 1. Standardization is less affected by outliers than min-max scaling, but neither method removes them.

The scaling parameters must be computed on the training set only and then applied unchanged to the validation and test sets. Computing them on all the data leaks information from the test set.

3. Encoding categorical features

Models need numbers, so categories must be encoded. One-hot encoding creates one binary column per category and works well when there are few categories. Ordinal encoding assigns integers and should only be used when the categories have a real order, such as small, medium and large. For high-cardinality features with thousands of values, target encoding or hashing is more practical, but target encoding must be computed inside cross-validation folds to avoid leakage.

4. Missing values

Missing values can be handled by deleting rows, by filling with the mean, median or most frequent value, or by model-based imputation. Adding an indicator column that marks which values were missing often helps, because the fact that a value is missing can itself carry information.

5. Creating new features

New features can come from combining existing ones, for example a ratio such as price per square metre, or from extracting parts of a date such as the day of the week or the month. Polynomial features, such as x squared or the product of two variables, let a linear model fit curved relationships. A log transform reduces the skew of heavily right-skewed variables like income.

6. Feature selection

Using too many features can cause overfitting and slow training. Filter methods rank features by a statistic such as correlation or mutual information. Wrapper methods such as recursive feature elimination train a model repeatedly with different subsets. Embedded methods such as L1 regularization (Lasso) shrink the weights of unhelpful features to exactly zero during training.